

#Aviation Meteorology



Chapter Two: Observed Weather Elements for Aviation



Objectives

- To be familiar with Surface winds for Aviation
- To know the Visibility and Runway Visual Range
- To be familiar with Clouds
- To be aware of Air temperature dew point temperature and Atmospheric Pressure



Surface winds

- It refers to the wind that blows at the earth's surface and can affect the takeoff, landing, and flying of aircraft.
- Wind is the horizontal motion of the air past a given point.
- Wind is measured in terms of velocity, a vector that includes direction and speed.
- The absence of apparent motion of the air is termed CALM.

- The direction and speed of the wind should be measured in an unsheltered, unobstructed area.
- Pilots need to know the direction and speed of surface winds to calculate their takeoff and landing distances and make adjustments to their flight plan accordingly.
- **Wind direction** is expressed in degrees, with the compass rose indicating the direction from which the wind is blowing.

- The wind speed is measured in knots, with one knot being equivalent to 1.15 miles per hour.
- The wind speed is typically measured using an anemometer.
- Surface winds can be influenced by a variety of factors, including local geography, temperature, pressure systems, and fronts.

- For example, winds blowing over mountains can create turbulence and updrafts, while winds blowing over flat terrain can create downdrafts and wind shear.

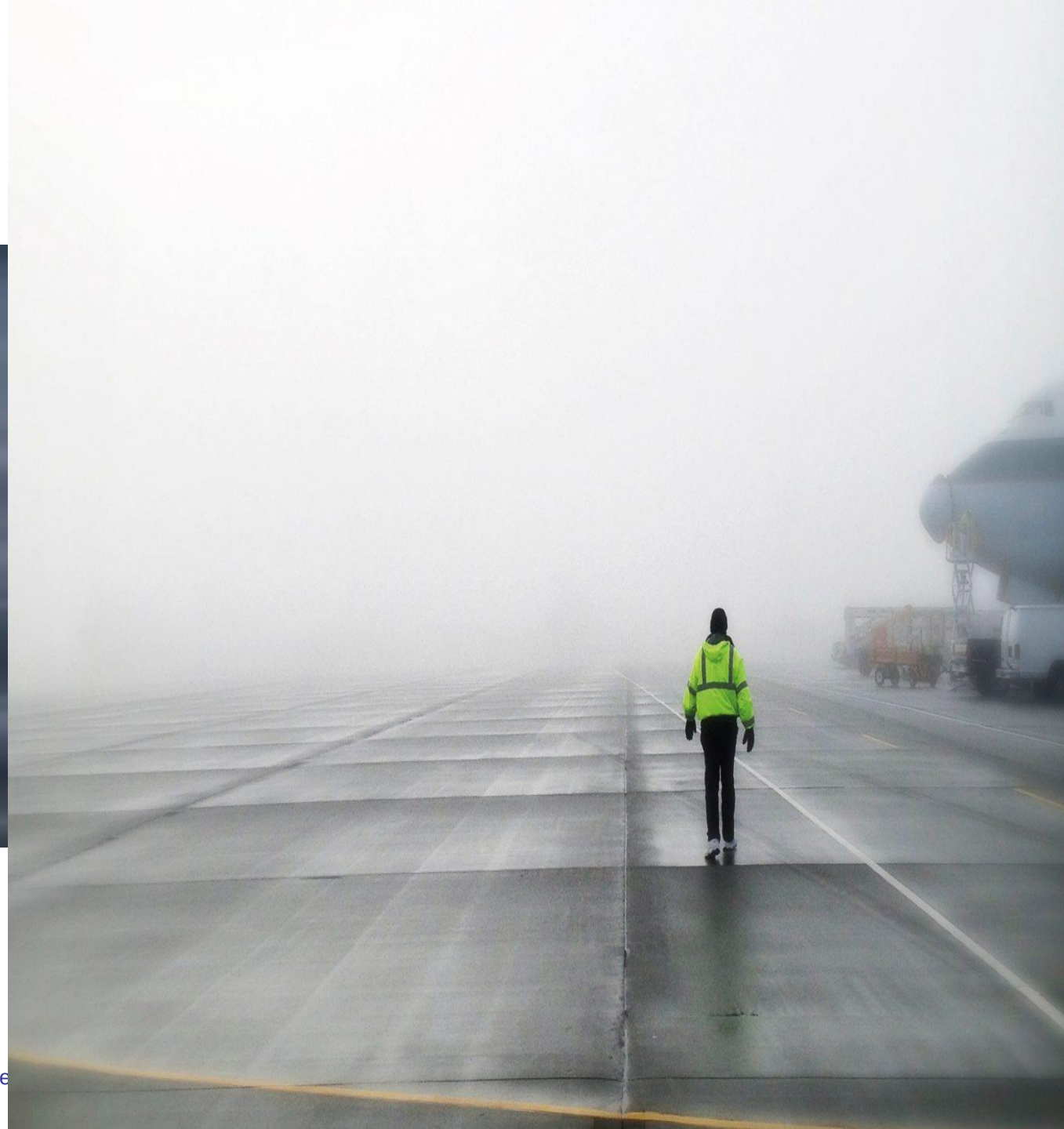
Visibility

- Visibility is a measure of the horizontal opacity of the atmosphere at the point of observation.
- Visibility expressed in terms of the horizontal distance at which a person should be able to see and identify specific objects.
- Visibility measured in meters or statute miles.
- In aviation, visibility is essential as it affects the safety of the aircraft and passengers.

- Poor visibility can make it difficult for pilots to see other aircraft, runways, or obstacles on the ground, which can lead to accidents.
- Visibility can be affected by a variety of factors, including fog, haze, smoke, precipitation, and darkness.
- In general, the lower the visibility, the higher the risk of accidents.

- For this reason, pilots are required to use instruments such as altimeters and navigational aids to maintain their flight path when visibility is poor.
- Visibility is typically reported at airports and other locations where aircraft take off and landing.
- The measurement is taken by a device called a transmissometer, which sends a beam of light through the air and measures the amount of light that reaches a receiver.

Visibility



Runway Visual Range (RVR)

- Runway Visual Range (RVR) is another crucial weather element observed in aviation.
- RVR is the distance over which a pilot can see the runway surface markings or lights.
- It is measured in meters or feet and is calculated using instruments that detect the amount of light that is reflected from the runway.

- RVR is particularly important during takeoff and landing as it helps pilots determine whether they can safely operate the aircraft.
- If the RVR is too low, it can make it difficult for the pilot to see the runway, which can lead to accidents.
- For this reason, RVR is reported at airports and other locations where aircraft take off and land.

- The RVR can be affected by a variety of factors, including fog, rain, snow, or dust.
- When the weather conditions are poor, the RVR can decrease, making it difficult for pilots to operate the aircraft safely.
- In some cases, the RVR may be below the minimum required for takeoff or landing, in which case the flight may be delayed or canceled.

- To measure the RVR, instruments called transmissometers or forward-scatter meters are used.
- These instruments emit light towards the runway, and the amount of light that is reflected back is measured.
- The RVR is then calculated based on the amount of light that is reflected.

Cloud

- A cloud is a visible accumulation of minute water droplets and/or ice particles in the atmosphere above the earth's surface.
- Clouds are a significant weather element for aviation as they can affect visibility and cause turbulence.
- Clouds are classified into different types based on their appearance, height, and composition.

- The amount of cloud cover is reported as a percentage of the sky covered by clouds.
- Cloud type, amount, height of the base, and tops are all important weather elements that pilots need to be aware of to plan their flight and avoid areas of potential turbulence.

Types of Clouds

TABLE 4.2 The Four Major Cloud Groups and Their Types

1. High clouds Cirrus (Ci) Cirrostratus (Cs) Cirrocumulus (Cc)	3. Low clouds Stratus (St) Stratocumulus (Sc) Nimbostratus (Ns)
2. Middle clouds Altostratus (As) Altocumulus (Ac)	4. Clouds with vertical development Cumulus (Cu) Cumulonimbus (Cb)

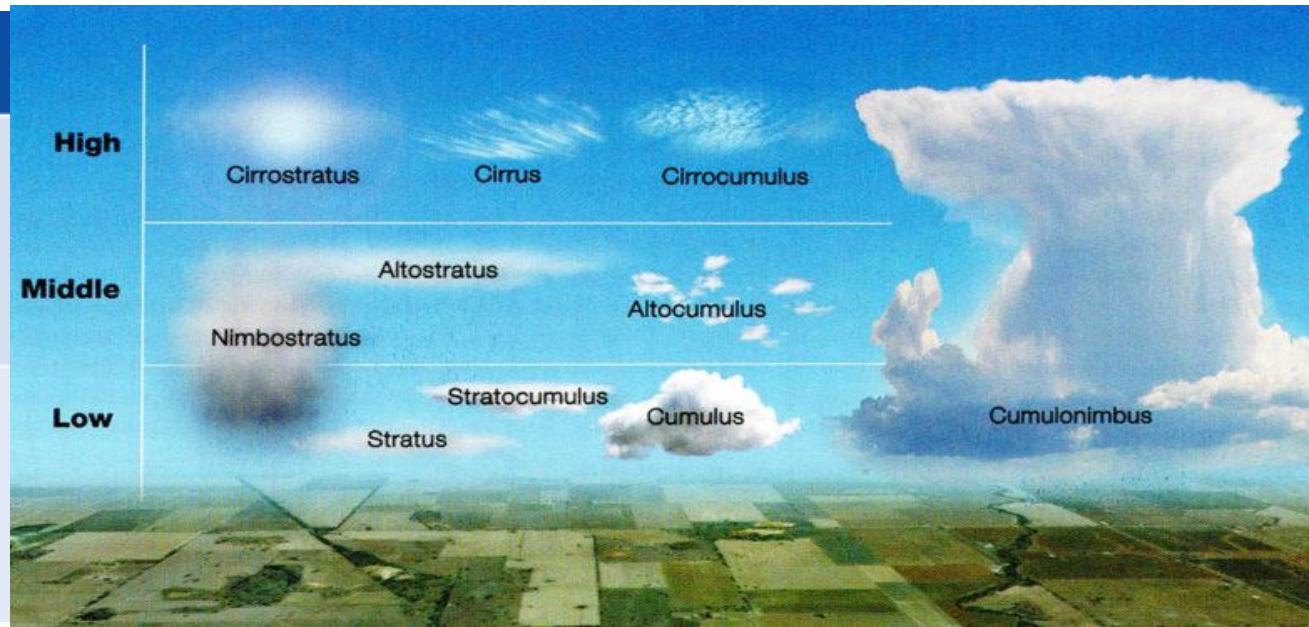


TABLE 4.3 Approximate Height of Cloud Bases above the Surface for Various Locations

CLOUD GROUP	TROPICAL REGION	MIDDLE-LATITUDE REGION	POLAR REGION
High Ci, Cs, Cc	20,000 to 60,000 ft (6000 to 18,000 m)	16,000 to 43,000 ft (5000 to 13,000 m)	10,000 to 26,000 ft (3000 to 8000 m)
Middle As, Ac	6500 to 26,000 ft (2000 to 8000 m)	6500 to 23,000 ft (2000 to 7000 m)	6500 to 13,000 ft (2000 to 4000 m)
Low St, Sc, Ns	surface to 6500 ft (0 to 2000 m)	surface to 6500 ft (0 to 2000 m)	surface to 6500 ft (0 to 2000 m)

Types of Clouds

- Clouds are classified into different types based on their appearance, height, and composition.
- The three main types of clouds are cumulus, stratus, and cirrus.

- ❑ **Cumulus clouds** are large and puffy with a flat base and rounded tops.
- They are often associated with fair weather, but can also indicate the potential for thunderstorms if they grow taller and become cumulonimbus clouds.
- Cumulonimbus clouds are thunderstorm clouds that can produce strong winds, heavy rain, hail, and lightning.

- These clouds can be dangerous for aviation, as they can cause turbulence, icing, and reduced visibility.



Cumulus cloud



Cumulonimbus cloud

❑ **Stratus clouds** are flat and featureless, covering the sky like a blanket.

- ✓ They are often associated with overcast or foggy conditions, and can cause reduced visibility and low ceilings.
- ✓ Stratus clouds can also be present in fair weather conditions, such as during stable atmospheric conditions.

- These clouds are composed of water drop lets and—when the temperature becomes low enough— some ice crystals.



Cirrostratus cloud



Altostratus cloud

❑ **Cirrus clouds** are **thin**, wispy, and often found at *high altitudes* blown by high winds into long streamers called mares' tails. that they can look like a white, feathery patch with a faint wisp of a tail at one end.

- ✓ They are usually composed of ice crystals and are often associated with fair weather.
- ✓ However, they can also indicate the potential for a change in weather conditions, such as an approaching storm.

- Cirrus clouds usually move across the sky from west to east, indicating the prevailing winds at their elevation.



Cirrus clouds



Cirrocumulus clouds

Amounts of Clouds

- The amount of cloud cover is reported as a percentage of the sky covered by clouds.
- It is important for pilots to be aware of the amount of cloud cover to plan their flight and avoid areas of potential turbulence.
- If the cloud cover is too thick, it can affect visibility and make it difficult to navigate the aircraft.

➤ The amount of cloud cover is usually reported as clear, scattered, broken, or overcast.

Height of Clouds base

- The height of the base is the altitude at which the cloud starts to form.
- It is important for pilots to know the height of the cloud base to avoid flying too close to the ground or obstacles.
- The height of the cloud base is usually reported in feet above ground level.

Tops Of Clouds

- The tops of the clouds refer to the highest altitude at which the cloud extends.
- It is important for pilots to know the height of the cloud tops to avoid flying into areas of potential turbulence or icing.
- The height of the cloud tops is usually reported in feet above ground level.

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- In conclusion, cloud type, amount, height of the base, and tops are all important weather elements that pilots need to be aware of to plan their flight and ensure the safety of everyone on board.

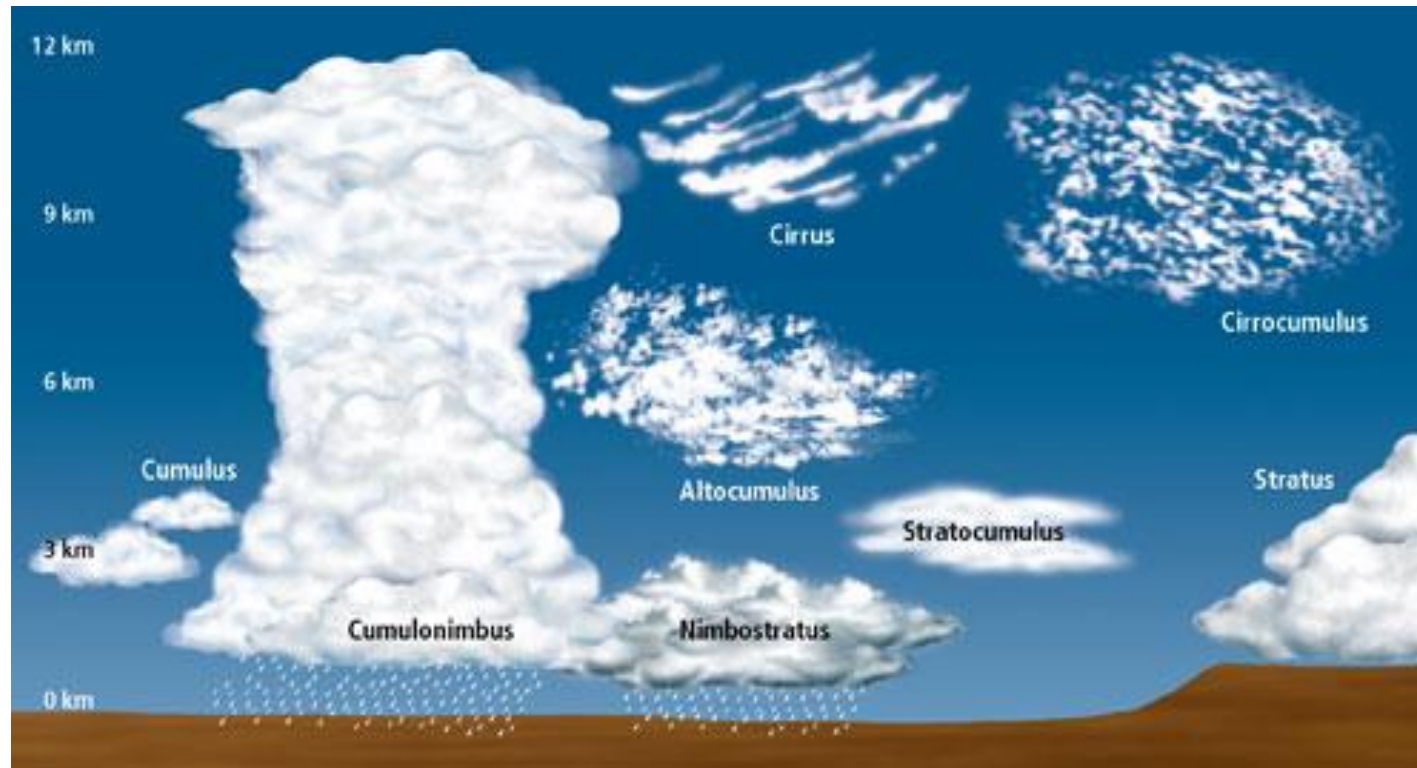


Fig: Different types of clouds at different altitude

Present weather

- **Present weather** refers to the current atmospheric conditions, such as rain, snow, fog, or thunderstorms.
- It is one of the critical weather elements observed in aviation as it can have a significant impact on flight safety.
- Pilots need to be aware of the present weather conditions to adjust their flight plan accordingly and ensure the safety of the aircraft and passengers.

- The presence of precipitation, such as rain or snow, can significantly reduce visibility, making it difficult for pilots to see other aircraft, runways, or obstacles on the ground.
- Precipitation can also create turbulence, which can cause discomfort for passengers and make it challenging for pilots to control the aircraft.
- To report present weather, meteorologists use codes that describe the type and intensity of the weather.

- For example, rain can be reported as light rain, moderate rain, or heavy rain, depending on the intensity of the precipitation.
- The codes are then transmitted to pilots through weather reports, such as METARs (Meteorological Aerodrome Reports), TAFs (Terminal Aerodrome Forecasts), and SIGMETs (Significant Meteorological Information).

- Accurate and up-to-date information about present weather is essential for pilots to make informed decisions about their flight plan and avoid potentially dangerous weather conditions.
- In conclusion, present weather is a critical weather element observed in aviation that can significantly impact flight safety.

Air Temperature

- Air temperature is the measure of the degree of hotness or coldness of the air.
- In aviation, it is measured in degrees Celsius or Fahrenheit.
- The air temperature affects the density of the air, which in turn affects the performance of the aircraft.
- As the air temperature increases, the density of the air decreases, which can impact the takeoff and landing distance of the aircraft.

- Pilots need to be aware of the air temperature to calculate their takeoff and landing distances accurately.
- Aviation is interested primarily in the temperature at engine intake levels above the runway but it is impossible to make regular temperature observations at such locations.
- Temperature readings are generally only required in whole degrees Celsius; corrections are applied before the readings are rounded off.

Dew Point Temperature

- Dew point temperature is the temperature at which the air must be cooled to reach saturation or 100% relative humidity.
- In aviation, it is essential because it can impact the formation of fog or clouds.
- If the air temperature and dew point temperature are close together, the likelihood of fog or low clouds forming is high.
- This can reduce visibility and make it challenging for pilots to land or takeoff.

Atmospheric Pressure

- **Atmospheric pressure** is the force per unit area exerted by the weight of the air above a given point.
- In aviation, it is measured in units of pressure, such as inches of mercury (in Hg) or millibar (mb).
- Atmospheric pressure is used to set the aircraft altimeter, which is an essential instrument for flight navigation.
- The altimeter measures the height above a reference point, such as sea level.

- Since the atmospheric pressure decreases with altitude, the altimeter needs to be set correctly to ensure that the aircraft's altitude is accurately measured.
- **Altimeter Setting (ALTSG):** Altimeter setting defines the pressure value to which an aircraft altimeter scale is set.
- Altimeter indicates the altitude above mean sea level of an aircraft on the ground at the location for which the value was determined.

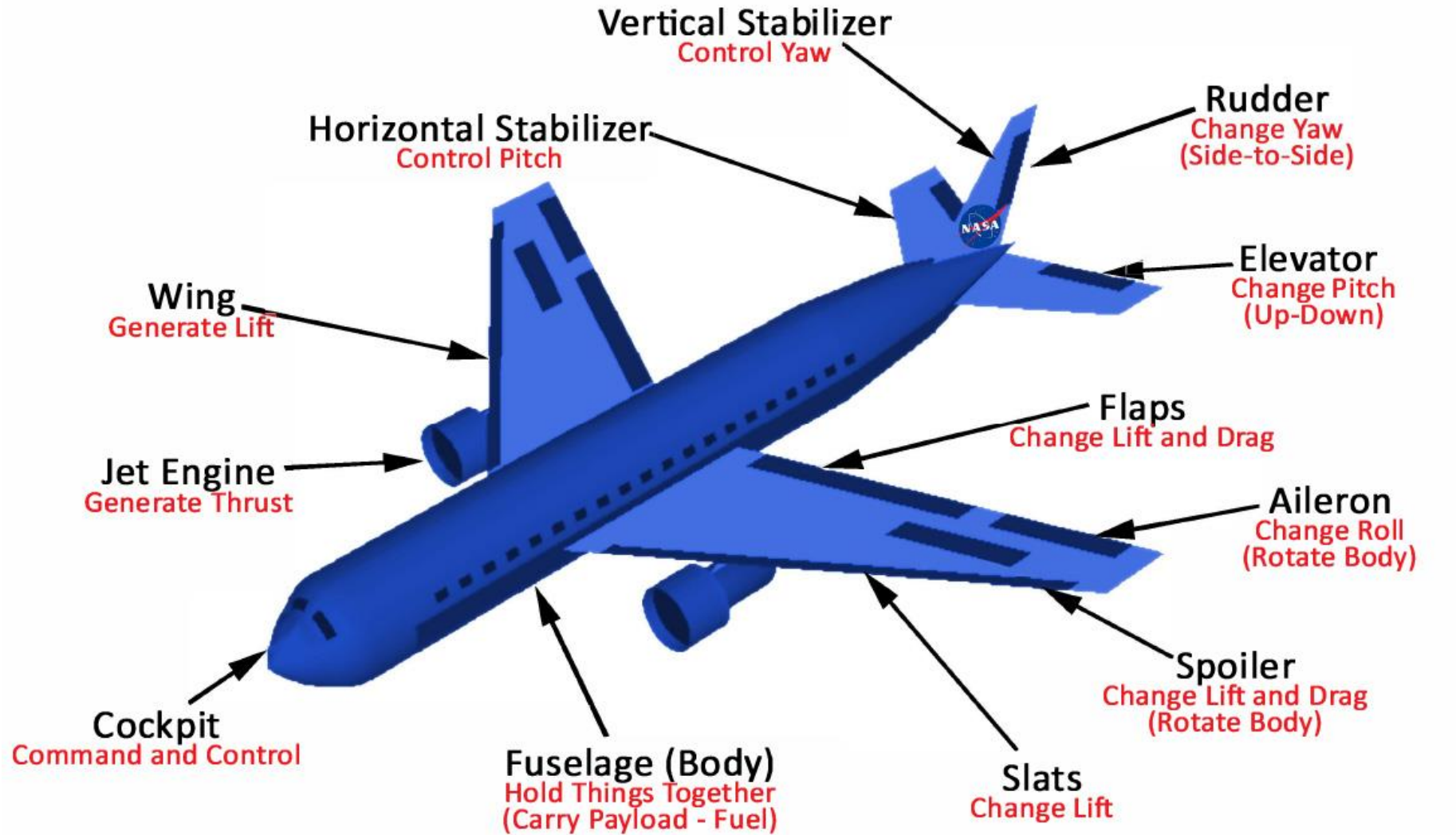
- Altimeter setting must be reported in the body of all reports (METAR and SPECI).
- **Barometric Pressure:** The atmospheric pressure measured by a barometer is barometric pressure.
- The term "barometric pressure" refers to the actual pressure sensor value.
- The sensor value may be an altimeter setting, station pressure, or simply a direct pressure value without applied corrections depending on the type of sensor.

- **Pressure Altitude, PA:** is the altitude in feet, in the standard atmosphere, at which a given pressure will be observed.
- It is the indicated altitude of a pressure altimeter at an altitude setting of 29.92 inches (1013.25 hPa) of mercury and is therefore the indicated altitude above or below the 29.92 inches constant-pressure surface.

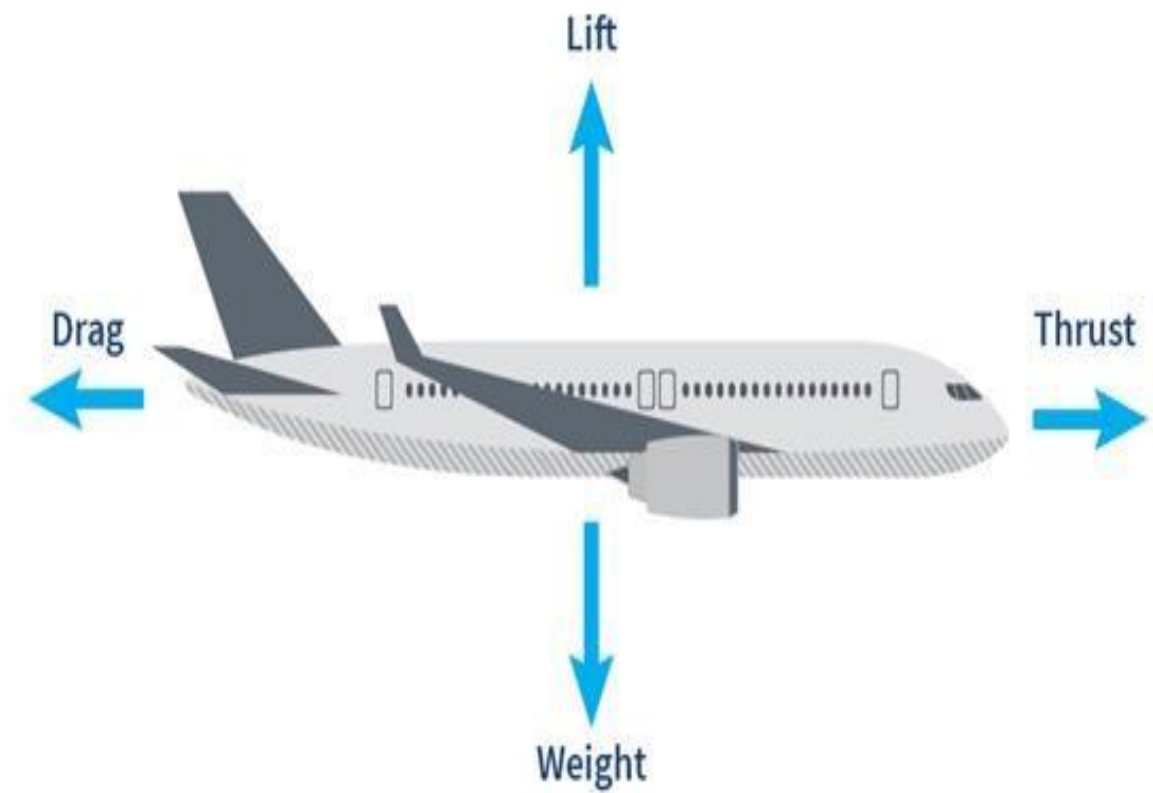
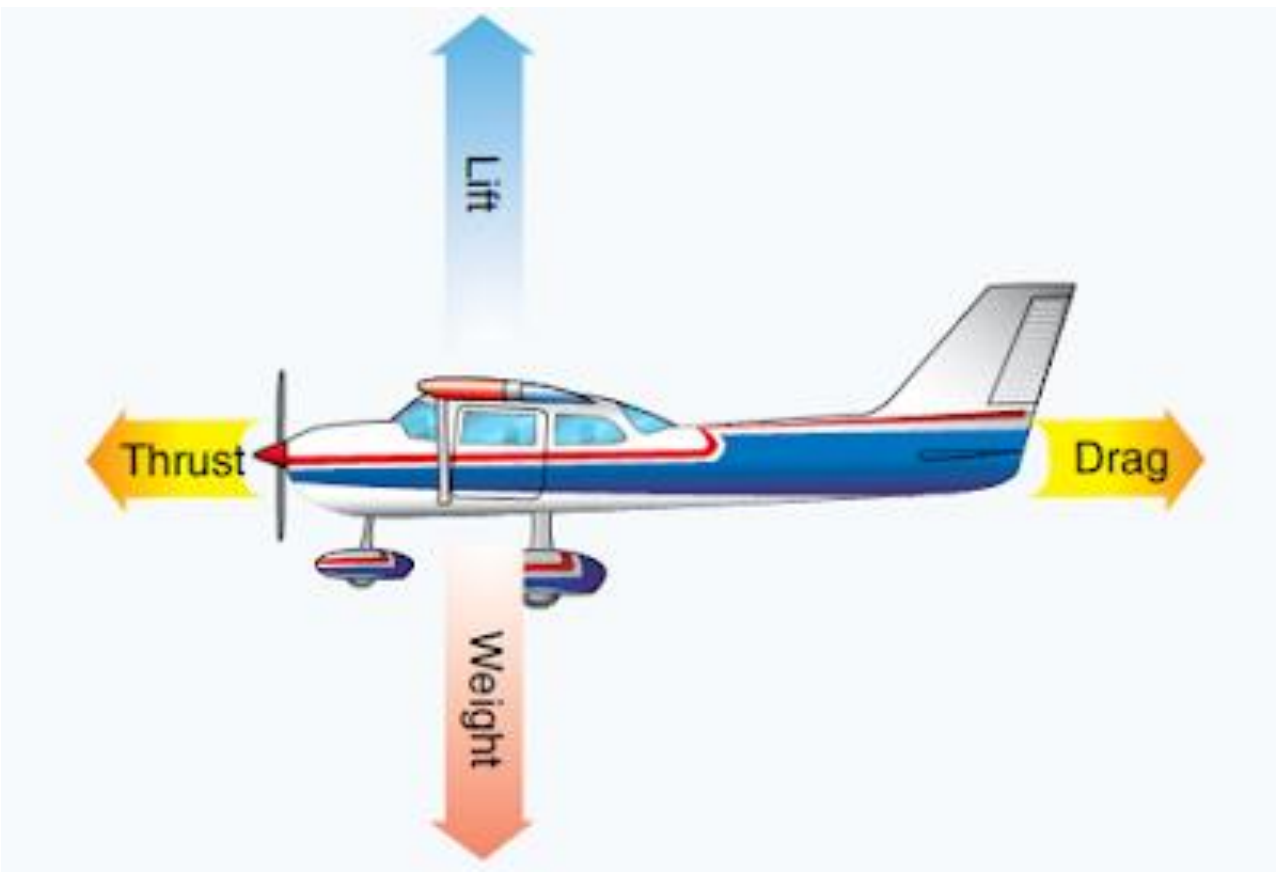
- **Pressure Change:** Pressure change is the net difference between the barometric pressure at the beginning and end of a specified interval of time, usually the 3-hour period preceding an observation.
- ***Generally, air temperature, dew point temperature, and atmospheric pressure are all critical weather elements that are essential for setting aircraft altimeters.***

- Pilots and air traffic controllers use this information to ensure accurate altitude readings and safe flight operations.
- Accurate and up-to-date information on these elements is necessary to make informed decisions regarding flight plans, calculate takeoff and landing distances, and ensure the safety of the aircraft and passengers.

Parts of Aircraft and their uses



Forces Acting On Aircraft



- Four primary forces are acting on an aircraft in flight: lift, weight, thrust, and drag.
- **lift** the upward acting force; **gravity**, the downward acting force; **thrust**, the forward acting force; and **drag**, the backward acting force (also called wind resistance).
- **Drag** and **gravity** are forces that act on anything lifted from the earth and moved through the air.
- **Thrust** and **lift** are artificially created forces used to overcome the forces of nature and enable an aircraft to fly.

❑ **Lift:** Lift is the force that opposes the aircraft's weight and keeps it in the air.

- The wings generate it as they move through the air.
- The shape of the wing, known as the airfoil, creates a difference in air pressure between the top and bottom surfaces of the wing.
- ***This pressure difference generates a force that lifts the aircraft into the air.***



Weight

□ Weight is the force of gravity acting on the aircraft.

- It is the force that opposes lift and is always acting downward.
- The weight of an aircraft includes the ***weight of the aircraft itself, fuel, cargo, and passengers.***

Thrust

- ❑ Thrust is the force that propels the aircraft forward.
 - ✓ ***Aircraft engine*** and propeller combination is designed to produce **thrust** to overcome drag
 - ✓ The amount of thrust generated depends on the power and efficiency of the engines.

Drag

- ❑ Drag is the force that opposes the aircraft's forward motion.
 - ✓ It is caused by the resistance of the air against the aircraft's surface and is influenced by factors such as the shape of the aircraft, the airspeed, and the altitude.
 - ✓ There are two types of drag: parasitic drag and induced drag.
 - ✓ Parasitic drag is caused by the friction of the air against the aircraft's surface, while induced drag is caused by the vortices that form at the wingtips as lift is generated.

- To take off and land, an aircraft must generate enough lift to overcome its weight and enough thrust to overcome drag.
- **During takeoff**, the pilot increases the thrust from the engines to reach a sufficient speed for the wings to generate enough lift to get the aircraft off the ground.

- **During landing**, the pilot reduces the thrust and changes the angle of attack to reduce the speed and generate enough lift to slow down the aircraft and bring it safely to the ground.
- Proper understanding and management of these forces is essential for safe and efficient aircraft.

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Thanks !